

Jaweria Fayyaz

✉ jaweriefayyaz474@gmail.com | ☎ 03320475160 | 🔗 linkedin.com/in/jaweria-fayyaz | 🌐 github.com/jaweriefayyaz

SUMMARY

AI/ML Engineer with hands-on experience in Machine Learning, Natural Language Processing, and Generative AI, combined with a strong foundation in data engineering on Azure Databricks. Skilled in building and deploying end-to-end ML/NLP pipelines, RAG-based applications, and scalable ETL workflows using PySpark and Delta Lake. Passionate about developing intelligent, data-driven systems that solve real-world problems.

EDUCATION

BS Computer Science

2021 – 2025

Govt. College University, Lahore — CGPA: 3.70

PROFESSIONAL EXPERIENCE

Associate Data Engineer

Aug 2025 – Mar 2026

Symufolk, Lahore

- Designed and maintained scalable batch and streaming data pipelines in Azure Databricks using PySpark, processing large-scale datasets for analytical and ML workloads.
- Performed data cleaning, validation, and feature preparation for downstream analytics and machine learning use cases.
- Optimized ETL workflows reducing average pipeline runtime by 25%, improving daily processing efficiency.
- Contributed to migration of legacy data from SharePoint into ADLS Gen2, building Bronze–Silver–Gold Medallion pipelines with Common Data Model harmonization.

PROJECTS

Anxiety Detection (Final Year Project) — [Link]

- Built an ML/NLP web app to detect anxiety presence, type, and severity from Urdu text; created a custom dataset of 12,000+ samples with tokenization, stopword removal, and TF-IDF preprocessing.
- Trained SVM, LightGBM, XGBoost, and XLM-RoBERTa; finalized DistilBERT achieving 91.31% accuracy; deployed via Flask (backend) and React (frontend).

Smart Resume Screener | RAG-Based HR Intelligence Tool — [Link]

- Built an end-to-end RAG pipeline using LangChain and FAISS to intelligently screen and rank multiple resumes based on natural language queries, simulating AI-powered HR recruitment.
- Implemented PDF ingestion, text chunking, and semantic embedding pipeline using HuggingFace sentence-transformers, storing vectors in a FAISS index for fast similarity retrieval.
- Integrated Gemini API for context-aware candidate comparison and ranking; deployed via Flask backend with React frontend for interactive resume querying.

Emotion Detection System — [Link]

- Built a real-time facial emotion classification system using DeepFace and OpenCV, categorizing expressions into 7 emotion types and 3 mood groups (Positive, Neutral, Negative).
- Implemented rolling prediction smoothing, confidence thresholding (40%+), FPS monitoring, and frame-skipping for optimized real-time performance.

SKILLS

Languages: Python, SQL, C++, Java

AI / ML & NLP: Machine Learning, Natural Language Processing, DistilBERT, XLM-RoBERTa, Scikit-learn, NLTK, Pandas, NumPy, Matplotlib, Seaborn

Generative AI: Retrieval-Augmented Generation (RAG), LangChain, FAISS, Vector Databases, Embeddings, Prompt Engineering, Gemini API

Data Engineering: ETL/ELT Pipelines, PySpark, Azure Databricks, ADLS Gen2, Delta Lake, Medallion Architecture, Azure Data Factory, dbt, Apache Airflow

Cloud Platforms: Microsoft Azure (ADF, ADLS Gen2, Synapse Analytics), Google Cloud Platform (BigQuery), Snowflake

Web Development: Flask, Django, React (Basic)

Databases: MS SQL Server, MySQL, PostgreSQL

Tools: Git, GitHub, VS Code, Jupyter, Google Colab, Azure DevOps